

$$|\Omega\rangle = \sum_k |k\rangle \otimes |k\rangle$$

We use the unnormalised EPR vector

$$|\Omega\rangle$$

. In coordinates, for $a, b \in n$,

$$|\Omega\rangle_{a,b} = 1$$

if $a = b$, and $|\Omega\rangle_{a,b} = 0$ otherwise.

Kronecker action on the EPR vector

Let $M, N \in (n \times n \text{ } \mathbb{C})$. Then

$$((M \otimes N)|\Omega\rangle)_{a,b} = \sum_{k,l} M_{a,k} N_{b,l} |\Omega\rangle_{k,l}.$$

Since $|\Omega\rangle$ is supported only on the diagonal $k = l$,

$$((M \otimes N)|\Omega\rangle)_{a,b} = \sum_k M_{a,k} N_{b,k}.$$

This is the lemma `kronecker_mulVec_epr`.

Therefore

$$(M \otimes N)|\Omega\rangle = 0$$

is equivalent to

$$\sum_k M_{a,k} N_{b,k} = 0 \quad \text{for all } a, b.$$

But

$$(MN^T)_{a,b} = \sum_k M_{a,k} (N^T)_{k,b} = \sum_k M_{a,k} N_{b,k}.$$

Hence

$$(M \otimes N)|\Omega\rangle = 0 \implies MN^T = 0.$$

Conversely, if $MN^T = 0$, then every coordinate above is zero, so

$$(M \otimes N)|\Omega\rangle = 0.$$

Thus

$$(M \otimes N)|\Omega\rangle = 0 \implies MN^T = 0.$$

This is the lemma `kronecker_mulVec_epr_eq_zero_iff`.

Alice-side injectivity

Take $N = I$. Then

$$(M \otimes I)|\Omega\rangle = 0 \implies MI^T = 0.$$

Since $I^T = I$,

$$(M \otimes I)|\Omega\rangle = 0 \implies M = 0.$$

In the Lean notation $M \otimes I$ is `bipartiteAliceLift M`. Thus

$$(\text{bipartiteAliceLift}(M))|\Omega\rangle = 0 \implies M = 0.$$

This is the lemma `alice_lift_mulVec_epr_eq_zero_iff`.

The affine Kronecker relation

Let $M, N \in (n \times n \mathbb{C})$. Assume

$$(I - M \otimes N)|\Omega\rangle = 0.$$

Then

$$|\Omega\rangle = (M \otimes N)|\Omega\rangle.$$

Taking the (a, b) coordinate gives

$$|\Omega\rangle_{a,b} = \sum_k M_{a,k} N_{b,k}.$$

The left hand side is $I_{a,b}$, and the right hand side is $(MN^T)_{a,b}$. Therefore

$$MN^T = I.$$

Conversely, if $MN^T = I$, then every coordinate of

$$(I - M \otimes N)|\Omega\rangle$$

is zero. Thus

$$(I - M \otimes N)|\Omega\rangle = 0 \implies MN^T = I.$$

This is the lemma `one_sub_kronecker_mulVec_epr_eq_zero_iff`.

With a scalar $c \in \mathbb{C}$, the same argument gives

$$(I - c(M \otimes N))|\Omega\rangle = 0 \implies c(MN^T) = I.$$

This is the lemma `one_sub_smul_kronecker_mulVec_epr_eq_zero_iff`.

On Alice's side only,

$$(I - c(M \otimes I))|\Omega\rangle = 0 \implies I - cM = 0.$$

This is the lemma `alice_lift_one_sub_smul_mulVec_epr_eq_zero_iff`.

Bipartite lift algebra

The bipartite lifts are

$$A_A = A \otimes I, \quad B_B = I \otimes B.$$

Their products reduce to ordinary Kronecker products:

$$(I \otimes B)(A \otimes I) = A \otimes B,$$

$$(A \otimes I)(I \otimes B) = A \otimes B,$$

$$(A \otimes I)(C \otimes I) = (AC) \otimes I.$$

Also,

$$(I - cA) \otimes I = I - c(A \otimes I).$$

These are the lift lemmas used before applying the EPR identities in Stage 3.

Stage 2 transpose and adjoint argument

Stage 2 uses positivity:

$$(T_1^2 + T_2^2 + T_3^2)|\Omega\rangle = 0$$

and the self-adjointness assumptions

$$T_i^\dagger = T_i.$$

Taking the inner product with $\langle\Omega|$ gives

$$\langle\Omega|(T_1^2 + T_2^2 + T_3^2)|\Omega\rangle = 0.$$

By linearity,

$$\langle\Omega|T_1^2|\Omega\rangle + \langle\Omega|T_2^2|\Omega\rangle + \langle\Omega|T_3^2|\Omega\rangle = 0.$$

Since $T_i^\dagger = T_i$,

$$\langle\Omega|T_i^2|\Omega\rangle = \langle\Omega|T_i^\dagger T_i|\Omega\rangle = \langle T_i \Omega | T_i \Omega \rangle = \|T_i \Omega\|^2.$$

Hence every summand is nonnegative, and all must vanish:

$$T_1|\Omega\rangle = 0, \quad T_2|\Omega\rangle = 0, \quad T_3|\Omega\rangle = 0.$$

The Lean proof must first show that the three concrete SOS terms are self-adjoint. The transpose-adjoint facts used are:

$$(XY)^\dagger = Y^\dagger X^\dagger,$$

$$(X - cY)^\dagger = X^\dagger - \bar{c}Y^\dagger,$$

$$I^\dagger = I.$$

For a finite product of commuting self-adjoint matrices,

$$\left(\prod_x F_x\right)^\dagger = \prod_x F_x^\dagger = \prod_x F_x.$$

This is the lemma `noncommProd_conjTranspose_eq_self`. Applied to Alice's row product,

$$\text{Row}_{i(A)}^\dagger = \text{Row}_{i(A)}.$$

The LCS sign is real:

$$\overline{(-1)^{b_i}} = (-1)^{b_i}.$$

Therefore, if $T^\dagger = T$ and $c = (-1)^{b_i}$, then

$$(I - cT)^\dagger = I - cT.$$

This is the helper `one_sub_smul_conjTranspose_eq_self`.

For the consistency term,

$$T_1 = I - B_j A_{i,j}.$$

Using $A_{i,j}^\dagger = A_{i,j}$, $B_j^\dagger = B_j$, and $A_{i,j} B_j = B_j A_{i,j}$,

$$T_1^\dagger = I - (B_j A_{i,j})^\dagger = I - A_{i,j} B_j = I - B_j A_{i,j} = T_1.$$

For the row term,

$$T_2 = I - (-1)^{b_i} \text{Row}_{i(A)}.$$

Since the sign is real and $\text{Row}_{i(A)}^\dagger = \text{Row}_{i(A)}$,

$$T_2^\dagger = T_2.$$

For the product term,

$$T_3 = I - (-1)^{b_i} \text{Row}_{i(A)} A_{i,j} B_j.$$

The core product is self-adjoint because the factors are self-adjoint and the local commutation rules reorder the reversed adjoint product:

$$(\text{Row}_{i(A)} A_{i,j} B_j)^\dagger = B_j A_{i,j} \text{Row}_{i(A)} = \text{Row}_{i(A)} A_{i,j} B_j.$$

Together with the real sign,

$$T_3^\dagger = T_3.$$

This is the transpose argument needed by Stage 2.

Stage 3 extraction

Let

$$A_{i,j} = A \otimes I, \quad B_j = I \otimes B, \quad \text{Row}_{i(A)} = \text{Row} \otimes I.$$

The three SOS terms are

$$T_1 = I - B_j A_{i,j},$$

$$T_2 = I - (-1)^{b_i} \text{Row}_{i(A)},$$

$$T_3 = I - (-1)^{b_i} \text{Row}_{i(A)} A_{i,j} B_j.$$

Assume Stage 2 has proved

$$T_1|\Omega\rangle = 0, \quad T_2|\Omega\rangle = 0, \quad T_3|\Omega\rangle = 0.$$

For consistency,

$$B_j A_{i,j} = (I \otimes B)(A \otimes I) = A \otimes B.$$

Thus

$$(I - A \otimes B)|\Omega\rangle = 0.$$

By the affine Kronecker relation,

$$AB^T = I.$$

If B is an observable, then $B^2 = I$. Transposing gives

$$(B^T)^2 = I.$$

Therefore

$$A = AI = A(B^T B^T) = (AB^T)B^T = B^T.$$

This is the lemma consistency_of_epr_annihilates.

For the row relation, set $c = (-1)^{b_i}$. Since

$$T_2 = I - c(\text{Row} \otimes I),$$

Alice-side injectivity gives

$$I - c \text{Row} = 0.$$

So

$$c \text{Row} = I.$$

Since $c^2 = 1$,

$$\text{Row} = cI = (-1)^{b_i} I.$$

This is the lemma row_relation_of_epr_annihilates.

For the product relation,

$$\text{Row}_{i(A)} A_{i,j} B_j = ((\text{Row } A) \otimes I)(I \otimes B) = (\text{Row } A) \otimes B.$$

Thus

$$(I - c((\text{Row } A) \otimes B))|\Omega\rangle = 0.$$

By the scalar affine Kronecker relation,

$$c((\text{Row } A)B^T) = I.$$

Equivalently,

$$(-1)^{b_i} (\text{Row } AB^T) = I.$$

This is the lemma product_relation_of_epr_annihilates.

Bundling the three extracted identities gives

$$A = B^T, \quad \text{Row} = (-1)^{b_i} I, \quad (-1)^{b_i} \text{Row } AB^T = I.$$

This is local_matrix_identities_of_sos_terms_annihilate_epr.